# Automatic Category Suggestions for Wikimedia Commons

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## 1. Our understanding of the problem

Wikimedia Commons contains a huge collection of freely usable media, but choosing the right categories for an uploaded image can take time. Our understanding is that the project should help users find relevant category suggestions automatically, so images can be organised more easily. The suggestions should be useful and, wherever possible, point to real categories that already exist on Wikimedia Commons.

## 2. Our approach

We are building the system around a pretrained CLIP image–text model instead of training a new model from scratch. CLIP can represent an image and category names in a shared embedding space, allowing us to compare them and rank candidate categories by similarity. To make the suggestions specific to Wikimedia Commons, we are collecting category names from the Commons API rather than relying only on a small hand-written list.

> **Planned flow:** user uploads an image → backend validates it → CLIP represents the image → the system compares it with stored category embeddings → results are ranked → the app returns category names and verified Commons links where available.

## 3. Why we chose this approach

We have limited time and ordinary laptop hardware, with no dedicated GPU. A pretrained model gives us a practical starting point without the cost and time of training from scratch. We plan to keep the model loaded and reuse category embeddings so the system does not repeat expensive setup work for every image. We will only consider fine-tuning if testing shows a clear need and there is enough suitable labelled data and time.

## 4. Work completed so far

| Area | Current progress |
|---|---|
| Pretrained model | A CLIP ViT-B/32 component has been implemented. Earlier test output reported that the model loaded and model-logic tests passed; the component exposes image/text embedding and label-scoring functions. |
| Environment | A project virtual environment is set up. FastAPI, Uvicorn and HTTPX were reported installed successfully; the project uses the virtual environment's Python executable because PowerShell does not recognise the short `python`/`py` commands. |
| Commons category collection | A category-tree walk reportedly collected 4,926 real Commons category titles. Some API requests encountered HTTP 429 rate limiting, so the collection/index needs validation. |
| Category index | The CLIP embedding index is currently being built from the collected titles. The latest status indicated the build had been stopped previously and was being retried. A successful saved index and real-image lookup have not yet been confirmed. |
| Backend / UI | The FastAPI backend and frontend integration are not yet confirmed complete. They still need to be connected and tested together. |

## 5. Where we are right now

We are at the category-index stage. The category names have been collected, and the next technical checkpoint is to finish encoding them with CLIP, save the index, reload it, and test a real image against it. We will not treat the

index as complete until these checks pass. The backend must then expose the image-to-category workflow through an API, followed by frontend integration and end-to-end testing.

## 6. What remains

• Finish and verify the saved category index; confirm the number of indexed categories and embedding dimensions.

• Complete the FastAPI endpoint that accepts an image and returns ranked category suggestions.

• Connect category retrieval to the backend and return canonical Commons links only when verified.

• Integrate the frontend with the agreed API contract.

• Test with representative images, review the quality of the top suggestions, and document limitations.

• Prepare setup instructions and a short demo flow.

## 7. How we will evaluate the result

We will test the full path from image upload to category suggestions using several visually different images. We will check that the API responds correctly, the returned categories are real where verification is claimed, and the ranking is useful. Where reliable ground-truth categories are available, we can calculate metrics such as Precision@5; otherwise, we will clearly label the review as qualitative. Similarity scores will be treated as ranking signals, not calibrated probabilities.

## 8. Current limitations and risks

The category index is not yet verified, and Commons API rate limiting can slow collection. CLIP ranks the candidates it is given; it does not guarantee that the best possible category is present in the candidate list. The final quality and CPU response time still need to be measured. At this point, we are reporting implementation progress, not claiming that the complete application is finished.

## Summary

We have chosen a practical, pretrained-model approach suited to our time and hardware constraints. The model component and initial Commons category collection are in place according to the current development logs. Our immediate priority is to verify the category index, complete the backend connection, and then test the full application before the final demo.

Status note: This report reflects the latest development status visible in our working session. Items described as “reported” or “in progress” should not be presented as fully verified until the corresponding tests have passed.